CREATING A COLLABORATIVE ACTION:
BENEFITS AND BARRIERS IN MULTIDISCIPLINARY DESIGN PROCESSES FOR SUSTAINABILITY

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ABSTRACT

In sustainable design multidisciplinary collaboration is often a necessity. In this kind of collaboration participants merge and translate their professional knowledge, requiring certain skills and proper management of the process. Earlier research on such collaboration identifies a transdisciplinary process to merge the different professional inputs towards a common aim, knowledge and value systems. It also outlines different psychological and professional approaches towards the collaborative, inter-professional design process.

This article studies literature on inter-professional design collaboration, and is reflecting the findings on data gathered with student questionnaires and professors' interviews from a multidisciplinary Master's programme in sustainable design. The aim is to study different approaches towards a shared, transdisciplinary design process between several professionals, and to better understand how to manage this kind of design collaboration.

Keywords: sustainable design; inter-professional design process; transdisciplinary design framework;

INTRODUCTION

The challenges of sustainable design - similarly to many other contemporary design problems - are often complex and multi-faceted, and their solutions will require mediation between several concepts, contexts and interests. They are the "wicked problem[s] for design in the twenty-first century" (Wahl, 2006). Sustainable design requires transdisciplinarity, in which different professional approaches are oriented towards a common objective with shared understanding of the problem, within a complex real-life problem context. Inter-professional collaboration together with expanding stakeholder participation is creating a new role for designer to work as the facilitator in a multidisciplinary co-creation process. The design practice can help to create more sustainable solutions by helping to reconcile different interests and values together and by merging knowledge between the stakeholders. This, however, requires a better design of the design process itself.

This article studies these topics through an extensive literature study, and reflects its findings against data that is gathered from a multidisciplinary Master's programme in sustainable design in Aalto University, Finland. The aim is to identify what aspects and processes should be emphasized in setting up inter-professional collaboration for sustainability.

TRANSDISCIPLINARY SUSTAINABILITY

Research on sustainability must inevitably have a trans-scientific character (Tukker et al. 2008). Sustainable design is merging values from several domains of science, societal groups and stakeholders, being fundamentally based on a transdisciplinary design dialogue (Wahl & Baxter, 2008). It is a process of coevolution and co-design (ibid.: 72). When collaborators aim towards a common goal, and their knowledge integrate further, more traditional multidisciplinary approach evolves into interdisciplinarity, and eventually, after this collaboration is linked to real-life cases and problems, into transdisciplinarity (Hukkinen, 2008).

Approaches to inter-professional design collaboration must embrace transdisciplinary thinking, not only in order to understand different professional competences and to utilize these in design outcomes, but foremostly to manage the design process itself. In addition to professionals' different types of knowledge frameworks and communication tools, also several world-views clash together. Different types of collaborators have perspectives to sustainability on different scales and systemic levels, and therefore their approaches should be properly balanced to better support the creation of successful transdisciplinary outcomes.

RESEARCH CASE AND AIMS

This conference paper makes an attempt to identify different types of collaborators, and the tools and frameworks that could help designers to better facilitate inter-professional collaboration. It also aims to create a better understanding of the balance between the discipline-centered and the holistic, shared processes. The focus in the interpretation is on identifying the different disciplinary and psychological competences and barriers in a collaborative, inter-professional design process in the context of sustainability.

This article studies the following questions:

- What are the different approaches to inter-professional collaboration, and what barriers they may cause?
- What types of skills, frameworks and tools are necessary in this type of collaboration?

The research material for this article is gathered from a multidisciplinary Master's programme called Creative Sustainability (CS), which initiated in Autumn 2010 in Aalto University, Finland. As a whole, the data consists of professor interviews and student questionnaires with qualitative and quantitative data. The quantitative assessment of the results is used mainly to identify, visualize and support the findings in qualitative data.

THEORETICAL BACKGROUND AND LITERATURE

It is commonly agreed that basic disciplines "cannot in isolation provide sufficient and necessary solutions for sustainability" (Shin et al. 2008: 1833). Instead, sustainable design requires widespread participation among stakeholders (Wahl & Baxter, 2008: 72), also extending outwards towards various representatives from several professional areas. This need calls also for ways to integrate several professionals' knowledge into a single successful outcome.

INTEGRATING MULTIDISCIPLINARY KNOWLEDGE

In such integration process the profession-specific knowledge has to be shared, rearranged and synthesized, resulting into new blended knowledge that is qualitatively different from its partial inputs (see Hukkinen, 2008). The sources of knowledge are still heterogeneous, deriving from the natural sciences, social sciences, lay actors and various sectors of stakeholders.

Knowledge in transdisciplinary design collaboration resembles "Mode 2" knowledge (Gibbons et al. 1994), which transcends from the "old paradigm of scientific enquiry" (i.e. "Mode 1") characterized by "internally-driven taxonomy of disciplines" and "autonomy of scientists" (Nowotny et al. 2003: 179). Mode 2 knowledge is socially distributed, application-oriented and transdisciplinary (ibid.). It is not necessarily "derived from pre-existing disciplines" but is generated within the context of application (ibid.: 186). It requires unconstrained and instantaneous interaction, and is "highly reflexive" (ibid.: 187).

Besides the necessary professional knowledge, transdisciplinary collaboration – and the creation of Mode 2 type of emergent new knowledge – requires systems knowledge (analysis of complex questions), target knowledge (ability to define goals) and transformation knowledge (Wiesmann et al. 2008: 436). Thus, the process requires professional and collaborative abilities, and skills for both reflection and synthesizing.

TRANSDISCIPLINARY DESIGN DIALOGUE

In transdisciplinary dialogue different professionals integrate and translate their knowledge (Bruun et al. 2005) and create an interaction space, which can be called for example as "mediation space" (Després et al. 2004), "input space" (Hukkinen, 2008) or "trading zone" (Gibbons et al. 1994). The sharing and merging

of knowledge is a cognitive process that utilizes the use of analogies or pattern recognition. Each expert's input provides elements and relationships that are adopted in the construction of a new mental model (Hukkinen, 2008: 71). Hence, two partial domains of knowledge – partial mental models – meet and meld (ibid.). This process resembles the creation of "integral vision" by Wilber (2001; as in Wahl & Baxter, 2008), a "framework for understanding, acknowledging, and weaving together different perspectives and worldviews" (Wahl & Baxter, 2008: 72). A similar framework is required for creating an integrated understanding of a problem on both detailed and holistic levels.

This type of scaling and reflection provides a useful process to structure the transdisciplinary design dialogues, offering a framework for the mediation with value systems and knowledge. The transdisciplinary framework can help us to conceptualize how different value systems and different thought concepts change our experience of reality, affecting to "what and how we design" (Wahl & Baxter, 2008: 72), thus raising the question "of not only problem solution but problem choice" (Klein, 2004: 518). This moves the emphasis in research and design processes towards a more normative and socially responsible approach than what was perhaps considered appropriate in traditional science in the past.

BARRIERS AND COMPETENCES IN COLLABORATION

Coming to a mutual understanding is the most obvious challenge in creating a joint problem space. There exist several barriers towards the sharing of knowledge. Established disciplines may fear that quality of science could be eroded or that their autonomy would be decreased (Nowotny et al. 2003). Furthermore, experts can feel vulnerable when their competence is redefined and new evaluation criteria are suggested (Klein, 2004). Disciplinary theories that only apply in more restricted domains are also often "wrongly scaled up" to infer universal laws (Hukkinen, 2008: 54). These differences and conflicts may lead to an overall lack of commitment, understanding and action, thus resulting in weak outcomes of collaboration.

In addition to different approaches to scales and specificity of problems, it can in general be problematic to open disciplinary competences to outsiders. There exists a risk of losing professional focus (Hukkinen, 2008). Disciplinary identities and perspectives still remain as a prerequisite in order to be able to properly implement the transdisciplinary dialogue.

Lastly, there are problems in the management of multi-professional design collaboration. Problems in managing multi-professional design and research include reinforced hierarchies and assessment systems encouraging into cynical game-play or industry-style production of results (Nowotny et al. 2003). Inter-professional collaboration for sustainability requires sharing, mediation and adaptation of knowledge, oriented towards a common goal within a usually complex problem setting. Hence, it should be supported differently than the traditional disciplinary processes that are aimed into specific and identifiable subtasks.

THE TYPES OF COLLABORATORS

Different professionals are taught to approach sustainability with different emphases, but there also seems to exist distinct types of approaches to collaborative processes. Stereotypically the "social scientist consults the natural scientist about what to implement and the natural scientist consults the social scientist about how to implement" (Pohl, 2005: 1171) but also other, more psychological aspects may play an important role in collaboration.

An earlier study regarding sustainability-oriented research (Pohl, 2005) found out that despite of their background the researchers tended to be either an "Engaged Problem Solver" engaged in solving environmental problems, or then a "Detached Specialist" who provides expertise to those solving the problems (ibid.: 1170). Engaged Problem Solvers avoid discussing (or even refuse to discuss) a topic in an abstract way, while Detached Specialists can discuss things in a more abstract, generalized and context-free manner (ibid.).

Inter-professional design collaboration requires flexible management. To tackle specific problems a common aim must be created, and respect and trust

among participants must be built, and this requires skills, space and support from the managers of the process. According to the literature, there are hybrid experts (Hukkinen, 2008) or gatekeepers (Gloor, 2006), who can handle both professional and "detached" approaches. Collaborators with these types of skills may be able to become the necessary facilitators to produce successful outcomes.

DESIGNER AS A FACILITATOR

Designers have the potential to act as transdisciplinary integrators and facilitators (Wahl & Baxter, 2008). They are trained to facilitate collaboration between stakeholders in a creative way and could use these capabilities to achieve more than just the incremental results created by traditional eco-design strategies (Sherwin, 2004). Along with novel design approaches that emphasize stakeholder participation such as system design, co-design or collaboration in multidisciplinary contexts for sustainability, the design profession is taking a more proactive role within society. Designer as a facilitator should promote the transdisciplinary dialogue by providing supporting instruments, tools, frameworks and concepts, which would help to translate the professional knowledge and understanding, but also to balance the different inputs.

There is a need to rearrange a particular discipline's knowledge in such a way that transdisciplinarity can be achieved (Pohl, 2005). Design practice is rearranging its knowledge according to its focus context and stakeholder interests by default. It has several frameworks for participatory design processes, and these can be reinterpreted to better support the multidisciplinary value and knowledge co-creation.

DESIGNING THE COLLABORATIVE DESIGN PROCESS

The "upstream" of design activity (Wahl & Baxter, 2008: 74) – the metalevel of design – focuses on mediating "conscious awareness, value systems, worldviews", whereas the "downstream" represents artifacts, institutions and patterns of production (ibid.: 73). Transdisciplinary design process aims into higher contextual level that focuses on mediation between value systems and visions and on expanding its knowledge basis with collaboration.

Sustainable solutions to the "wicked problems of design" require, besides new ways of life and changes in meaning, also new design processes (Wahl & Baxter, 2008: 82). According to the findings from the literature this new, inter-professional design process framework should emphasize four different dimensions: 1) disciplinary competences, 2) skills to translate knowledge, 3) collaborative skills, and 4) management skills (see Fig. 1).

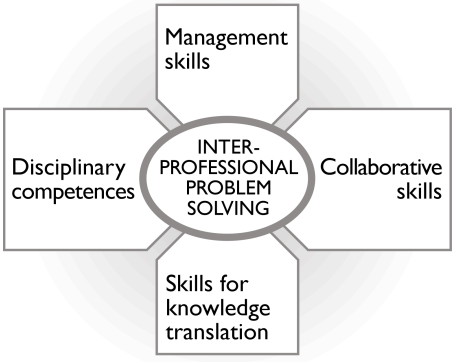


Figure 1. The setting for inter-professional problem solving

To further improve the transdisciplinarity in inter-professional, collaborative design projects, the disciplinary competences and the shared processes and tools have to be further studied, and their relations re-evaluated.

METHOD AND CASE

This article focuses particularly on the Creative Sustainability (CS) Master's degree programme in Aalto University, Finland. This particular focus has been selected as the aim of the programme is to foster sustainable innovation through principles of urban and industrial sustainability and corporate responsibility, and especially because the programme and its individual modules serve as a suitable laboratory for both initiating and observing transdisciplinary projects.

Data was gathered with questionnaires among the students of the CS programme (28, from two study modules). Additional data was acquired from an earlier study (Marttila & Kohtala, 2010), which analyzed interviews of CS programme professors (2 from design and 2 from business management).

FOCUS CASE AND ITS BACKGROUND

Creative Sustainability (CS) is a new and multidisciplinary Master's Programme in Aalto University, Finland, that begun first as a minor (2009) and then as a full Master's programme (2010). Students of the programme have their educational backgrounds (BA level) on various areas, and they are accomplishing their Master's degree in the fields of design, engineering, architecture, real estate, and business and management. The student respondents represent a first group of full MA students in the programme.

The programme's identified competence areas include strong professional knowledge, multidisciplinary approach, process management, design thinking and systemic approach (see http://www.creativesustainability.info/). All CS students have attended to two shared introduction courses on systemic approach (2 ECTS) and to a course on ethics of sustainability (2 ECTS). During the academic year (2010-2011) these students have also attended to complete modules (up to 15 ECTS) that were engaging several professions in multidisciplinary teamwork.

These students have been intentionally applying to "multidisciplinary" programme. This fact support the assumption that they represent a group that is open to multidisciplinary or even transdisciplinary processes and practices, and already in the beginning share some interests regarding sustainable design.

STUDENT QUESTIONNAIRE

A questionnaire was created based on the findings from the literature study and earlier research. It consisted of several rating scale questions, and of one open-ended question regarding benefits and barriers of the multi-professional collaboration. The first part of the questionnaire targeted to create visual support for the qualitative findings from the open-ended question. This part consisted of triangular grids that considered the balancing between the different dimensions of sustainability, and between the emphases in multidisciplinary collaboration (see Fig. 2).

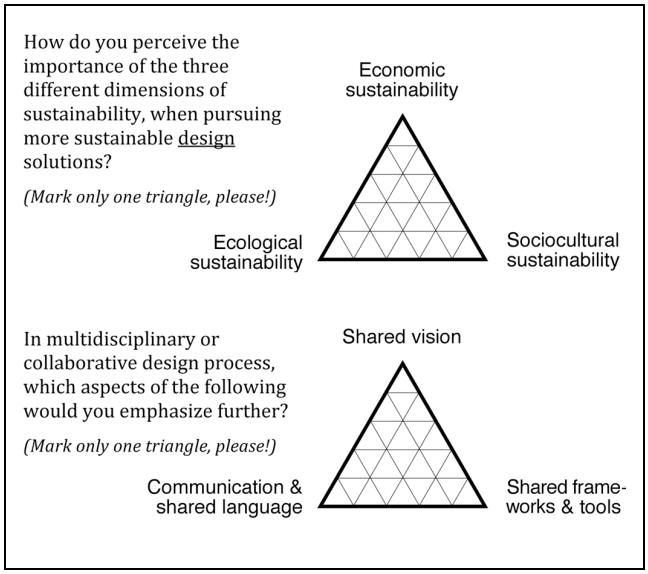


Figure 2. First part of the questionnaire (Author, 2010)

The second part of the questionnaire consisted of questions regarding multi-professional collaboration. In this part the first part's latter question was opened up in more detail, in comparisons between the shared and the disciplinary emphases. The general inquiry of the first part was elaborated further into three specific rating scale questions (see Fig. 3).

The third and final part of the questionnaire was an open question regarding multi-professional collaboration. Answers were collected to a simple question: *"What benefits or barriers were visible in the interaction"?*

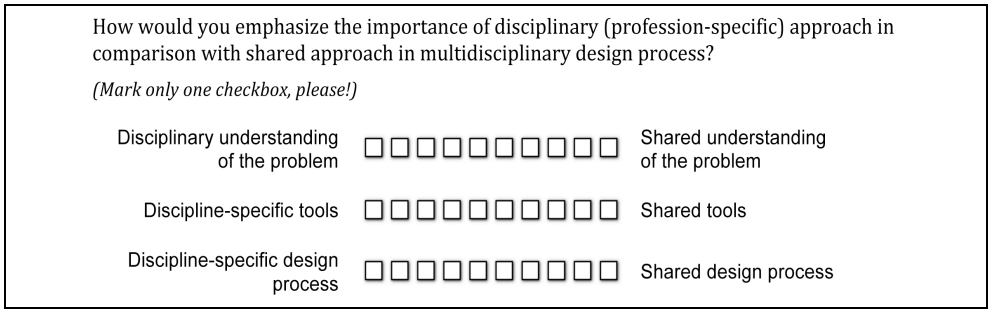


Figure 3. Second part of the questionnaire focusing to multi-professional design process (Author, 2010)

ANALYZING THE DATA

In the data analysis there exists some pre-assumptions gathered from the literature. Against previous studies it is justified to assume that transdisciplinary processes require a common objective and understanding. In general, a shared view of the problem is an essential thing, but achieving this requires certain experience of inter-professional knowledge building, strong professional know-how and proper management.

The idea in the ranking scale questions was to create visual support for the qualitative data. Unlike traditional Likert scaling, which is a bipolar scaling method that measures either positive or negative response to a statement with limited choices, the idea was to look for differences in how psychological types process their answers, and how extreme they are in their selections. Thus, the scale also forced into a choice between the options. However, the main focus remained on qualitative interpretation of the data that was gathered from the open-ended question in the questionnaire.

DATA AND RESULTS

Data was gathered during Autumn 2010 and Spring 2011 from two study modules in the multidisciplinary CS programme, another organized by Department of Design, and the other by Department of Architecture. These modules had students with backgrounds in various design fields, such as industrial (7), product (2), spatial (2) and graphic design (2), and in engineering (2), architecture (6), real estate (2), business and management (2) and media communication (1). Two of the answerers participated in both modules, but were included as two separate answers as the replies were gathered from different contexts. The total sample size is thus 28. Majority of the respondents were females (17 of 28) and many had foreign backgrounds (12 of 28).

Additional data and support was gathered from earlier interviews with professors (2 from design, 2 from business and management) that were implemented in 2010 (see Marttila & Kohtala, 2010). These professors were involved in CS progamme planning.

HOW TO APPROACH INTER-PROFESSIONAL DESIGN PROCESS?

In the earlier interviews all the CS professors agreed that the problems related to sustainability are complex and require collaborative action. The starting point was seen in initiating a dialogue between professions, and for this particular purpose real-life contexts and cases can offer the space where merging of the disciplinary aspects can happen (Marttila & Kohtala, 2010). CS professors were very eager to see *"decisions that would come out of very kind of dense interaction, and mutual understanding"*.

In sustainable design actions should be balanced within the three dimensions. Furthermore, mediation between the collaborators' different professional perspectives requires balancing between the disciplinary and the shared processes. This is visible in the centrally clustered results of the first part in the student questionnaire (see Fig. 4).

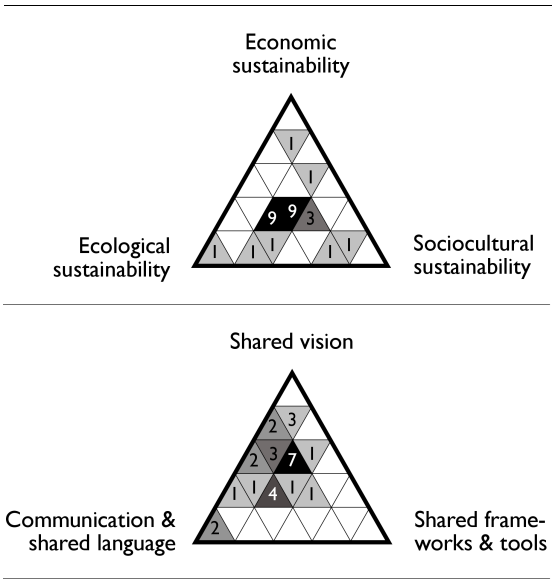


Figure 4. Results to first part of the questionnaire (n=28)

In the visual data from the latter triangle regarding "multidisciplinary collaboration", however, the emphasis seems to tilt towards the left side of the triangle and away from shared frameworks and tools. Shared vision, communication and language between the participants is seen as more important: *"[…] a vision should be shared in order to come up with a solution for a common problem"* (student).

"It was challenging but interesting to understand other [professional] languages and figure it out to way to put ideas together" (student).

In the second part of the questionnaire the students answered to three rating scale questions. In the figure below this data is clustered according to these questions, and the amount of answers is shown respectively on the scaling axis (see Fig. 5).

CHALLENGES AND BENEFITS IN THE INTER-PROFESSIONAL DESIGN COLLABORATION

Sustainable design requires transdisciplinary dialogue. The benefits of inter-professional collaboration include an increase in knowledge and expanded perspectives, but this requires specific understanding:

"[Y]ou also have to have an understanding of the (...) real processes that are in use currently, and then that are in the opportunity space that can be created through the combination of professions" (professor).

The challenge in multi-professional design collaboration is the fact that one has to also be able to understand the other professions, especially if there is a common aim. Emphasis is in understanding what king of input is needed, in how to describe own field knowledge, and how to know others' expertise:

"[Challenge is] learning to understand what other disciplines need as an input in order to contribute to the project work" (student).

"[One] would pay more attention to other field's knowledge and learn to describe own field knowledge to other background people" (student).

Such collaboration requires reflective and continuous dialogue, and this requires skills to manage participation, so that the common aim is sustained:

"Too much focus on output which makes you divide the tasks too easily. This do[es]n't allow enough dialogue and possibility to share world-view[…]" (student).

One teacher from the latter CS module (from the Department of Architecture) pointed out that most successful teams had members with some existing work experience. In real-life design projects they might have learned some of these skills, while practicing collaborative design.

ASSESSING THE DIFFERENT APPROACHES TO COLLABORATION

The respondents emphasized the importance of a shared understanding of the problem a bit more than disciplinary, supporting the preliminary assumptions, but not very strongly as the mean for all answers is 1.5 on a scale of -5 to 5. For the question regarding the understanding of the problem, the distribution of answers seems quite random, except for the two bigger pillars on the shared side. However, although the sample size is quite small, there seems to be a differentiation into two groups (see Fig. 6).

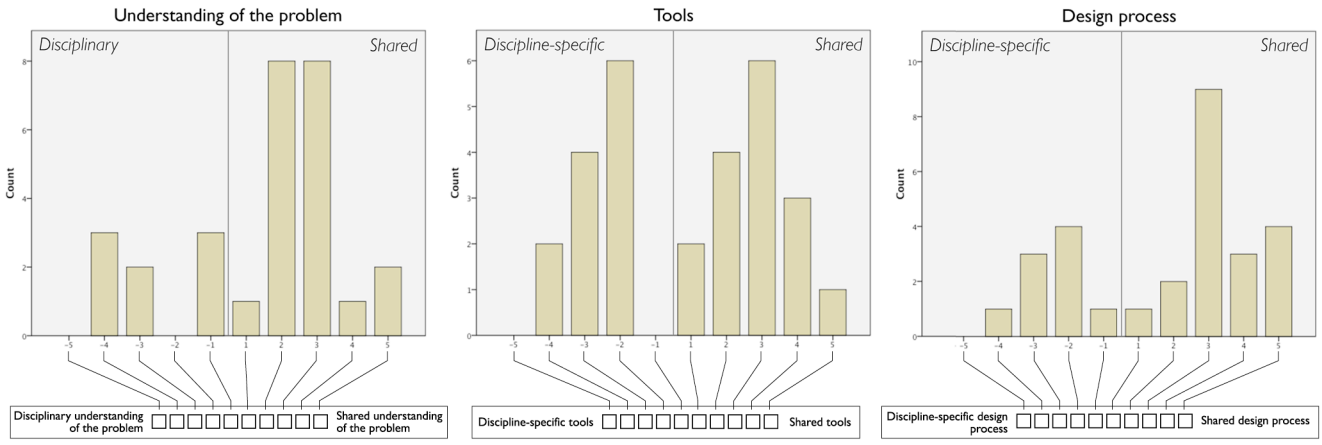


Figure 5. Results to the second part of the interview (n=28)

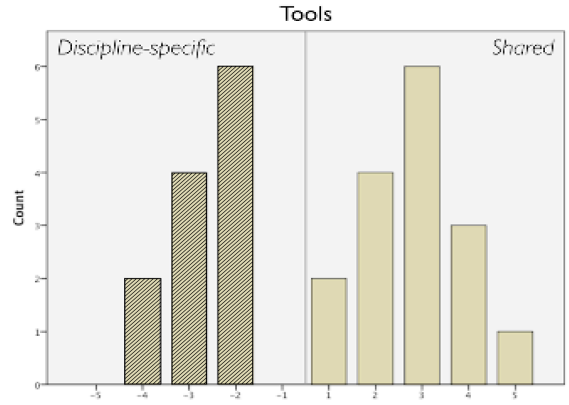


Figure 6. Repondents appreciating either discipline-specific tools (group 1: 12 of 28 replies - shaded) or shared tools (group 2: 16 of 28 replies - unshaded)

If the results are then separated in two groups according to the preference for discipline-centered or shared tools, according to the pre-assumptions the group that appreciates discipline-centered tools should also support discipline-specific focus in process, and maybe even shared understanding, identifying the distinct approach to collaborative design process. To some extent the results seem to support this assumption, and more strongly in relation to the process, which seems natural as tools are connected to the design process (see Fig. 7).

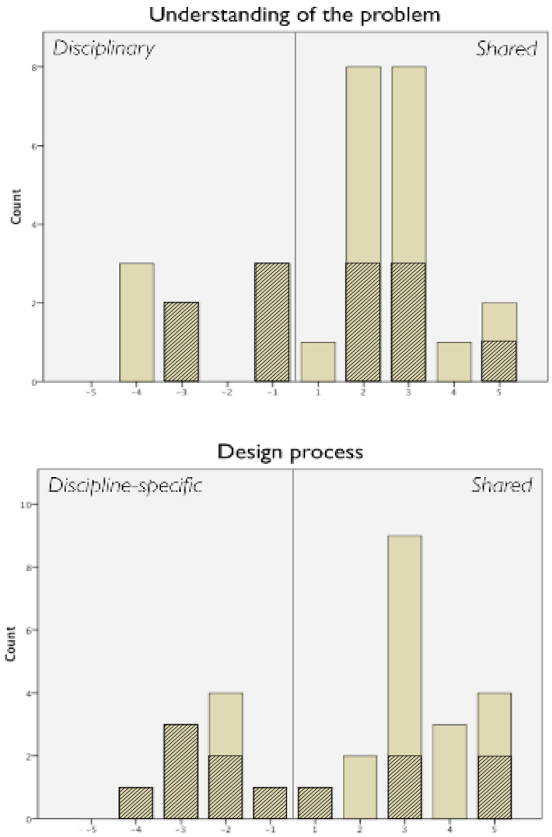


Figure 7. Distribution of the replies that emphasized discipline-specific tools (group 1 - shaded) over shared tools (group 2 - unshaded)

12 out of the 28 answerers preferred disciple-centered tools (see Fig. 6), and of them more than half preferred also discipline-centered process (see Fig. 7). Of the 16 answerers preferring shared tools, majority (14 out of 16) preferred shared process as well (see Fig. 7). Interestingly, however, the second group (unshaded) that is more oriented towards shared tools is also more clearly oriented towards shared process (see Fig. 7). This may suggest, that as one learns to participate into interdisciplinary design collaboration that shares understanding and goals, one also learns to better appreciate the shared tools and processes.

ANALYZING THE RESULTS

Although the sample size is small, the results suggest that preliminary assumptions were correct at least to some extent. In general, the experiences by the participants were defined positive, and both courses were quite successful in the end:

"It was a positive and interesting experience, how you can profit from other's knowledge and ideas" (student).

"Helps to approach issues in various perspectives. Beneficial to gain deeper knowledge and know-how. Resembles real world's professional interactions. (student)"

"This [multidisciplinary approach] improved the possibility of finding problems and solutions" (student).

While the results are not totally clear regarding different approaches to multidisciplinary design process, they offer preliminary suggestions for future design frameworks and tools. There exists some proof that persons emphasizing the importance of shared tools also emphasize the shared process, while the other group is not so clearly oriented. This suggests the importance for experience and experiencing:

"Initially it was challenging to see so different points of view, but with the time I became more flexible and learned to respect them" (student).

For those answerers that appreciated discipline-specific tools over the shared ones, emphasis seemed to be more on "learning" and "getting used to" collaborate, and on getting to know about others potential, as in the other group it was on learning about appreciating and respecting other "points of view", "world-views" or "field knowledge" and on sharing them. It seems that the latter group shows somewhat more mature approach to the collaboration.

According to the findings both shared and professional levels should be promoted to support more traditional approach in addition to the transdisciplinary reflection, and to prevent the inability to utilize every participants' specific expertise. Inter-professional, collaborative processes should therefore allow iterative and expanding learning on several levels of the problem.

Collaborative networks often suffer from the lack of effective and committed mediators that can facilitate the process, moderate the participation, and negotiate the power relations. These types of mediating skills can be learned only by experience:

"It is hard to know how to work in order to get the best result, if you are not used to working in multidisciplinary teams" (student).

"Finding conflicts between people with different backgrounds helps to learn a lot of things" (student).

Despite the challenges mentioned above, the multi-professional setting was perceived beneficial, and even the conflicts were perceived to benefit the participants.

DISCUSSION

This study identified the emergence of two main categories of practices within the group: Those who emphasized the "newness" of the experience, and those who were already used in inter-professional collaboration, and emphasized the importance of management. Furthermore this study identified two domains for activity, the disciplinary and "engaged" area and the holistic, "detached" area, where translation takes places. Interaction between these domains should stay constant and reflective, but well managed.

INTER-PROFESSIONAL DESIGN FOR SUSTAINABILITY

We may not be able to design "universally applicable blueprints" for sustainability (Wahl & Baxter, 2008: 73), but we may be able to develop the design process itself to support transdisciplinarity, since this helps to mediate more sustainable outcomes. In this process the application of dialogue and the gradual integration of knowledge in thinking and practice help to integrate diverse approaches into "a more inclusive basis" for complex decision-making process (ibid.: 77).

The expanding disciplinary approach affects to the professional identity of a designer. The collaboration with several professionals, experts and stakeholder networks creates a new role for design that is aimed at "promoting, facilitating and setting the conditions" for system innovation (Vezzoli et al. 2008: 2). In this process attention must be paid to build "value-consciousness" among researchers (Wiesmann et al. 2008: 438), to allow the negotiation between value systems. Such attitude requires new skills and emphasis on epistemological reflection. The type of new transdisciplinary knowledge must be "socially robust" to withstand the shift towards more open knowledge environments (Nowotny et al. 2003: 191).

MANAGING TRANSDISCIPLINARY COLLABORATION

The production of new knowledge, "however widely distributed, however trans-disciplinary, however heterogeneous, however reflexive" must be managed (Nowotny et al. 2003: 189). In the earlier studies regarding multi-professional environmental research most of the thinking about transdisciplinary collaboration was found to exist "at the level of programme management" (Pohl, 2005: 1159). Understanding the shared design process is similar to the process of learning to be a "hybrid expert" (Hukkinen, 2008), perhaps most necessary for the persons that try to facilitate and manage the process.

If the designer tries to take a facilitator's role in this type of design process, she has to be able to comprehend the benefits of multiple voices in the collaborative design process, and to manage the process with an understanding of the different professional views, of appropriate tools for epistemic translation, and of the collaborative frameworks and tools for managing such participation.

Followed by Pohl's typification into Detached Specialists and Engaged Problem Solvers (2005), the multi-professional design framework for sustainability should support both, "detached" processes (dialogical; integrating; value systems centered), and specific, "engaged" skills (profession specific; based into detailed concepts) as well as general skills for management and collaboration (see Fig. 8).

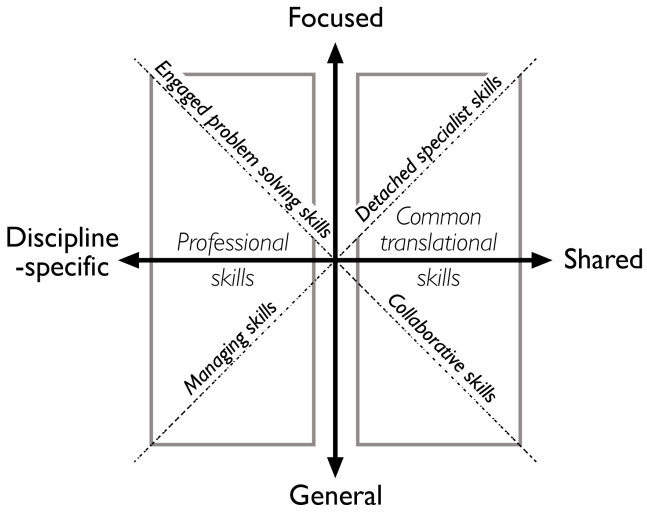


Figure 8. Skill types necessary for managing and participating into interdisciplinary collaboration (Author, 2011; based on Pohl, 2005)

The emphasis for multi-professional collaboration has to be on expanding the professional know-how, but in sustainable design the problems are often so complex that they require transdisciplinary approach. Therefore the participants should also be educated in transdisciplinarity. A cohesive and holistic approach enables disciplines "to better perform their proper functions" (Shin et al. 2008: 1834). Transdisciplinarity thus does not intend to damage the "disciplinary core", but to "rearrange" particular discipline's knowledge (Pohl, 2005: 1175), creating a new form of specialization (Wiesmann et al. 2008: 436).

NOT PROFESSIONAL ROLES, BUT PSYCHOLOGICAL APPROACHES

While the disciplinary approaches towards design process may vary from focused to more general, these approaches are not necessarily dictated by the participant's professional or personal characteristics, but may instead be learned through experience in earlier projects and work. Hence, the transdisciplinary processes can be learned. The emphasis seems to be on how the collaborative design process is managed, and how time and resources are shared between the participants.

The process to trade, merge and co-create knowledge might benefit from an assisting integrator. To some extent designers can be perceived to be involved in such process through their practice. Designers can help to change worldviews and value systems (Wahl & Baxter, 2008), either in the form of translating and scaling their professional knowledge and experience as well as others, or then in creating a positive account on transdisciplinary collaboration. Therefore also the general design framework for such process should support positive experience creation.

Further research should look into the tools for reflection and integration. Also, the focus could be on identifying how the participation into inter-professional collaboration develops the participants approach to multidisciplinary collaboration in general, by follow-up inquiries and in-depth student interviews.

CONCLUSION

In sustainable design the proper fostering of transdisciplinarity in collaboration can encourage participants "to contextualize and situate their specialist knowledge within a larger holistic/integral meta-perspective" (Wahl & Baxter, 2008: 74). This paper looked to further define the transdisciplinary design process and the differences between the disciplinary and shared knowledge creation processes in inter-professional design collaboration. The data was assessed to identify the ways participants approach shared understanding, design tools, frameworks and process.

The focus in managing and participating the inter-professional, multidisciplinary collaboration is on two distinct areas: Professional practices and shared synthetic processes. The interaction between these domains should be reflexive and continuous, aiming into expanding the scope of action towards holistic understanding of the problem at hand. Also the tools for collaboration should support the translation of distributed professional knowledge, focus on the application, and thus be based on transdisciplinary dialogue.

The transdisciplinary and collaborative design framework should aim to the upstream of design, assessing a problem integrally and within a joint problem space, and by merging value systems. In parallel with the iterative co-creation and orientation of the problem, there should be a professional loop, in which the focus is on specialist and profession-specific tools. This type of multi-level collaboration is necessary to allow the professional expertise to link with a mediated view of the problem, to set up the transdisciplinary dialogue, at the same time accomplishing specific tasks defined by the professional background.

Inter-professional, multidisciplinary design collaboration requires fluent management that creates possibilities for a variety of different types of participation, with different areas of expertise. Designers already have many competences to be the integrators and facilitators of such transdisciplinary collaboration, and they could use these abilities to implement better transdisciplinary dialogues, for more sustainable design outcomes.

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